AE4868 – Numerical Astrodynamics

Mission Design - Differential correction

Dominic Dirkx and Kevin Cowan

Differential Correction

Introduction

- We have two models
  - **Analytical** model - Lambert arc
  - **Numerical** model - Tudat propagation

- Analytical model
  - Exactly meets initial and final position constraints
  - Only valid under very basic dynamical assumptions (see previous video)

Differential Correction

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  - Initial state provided by analytical model
  - Does not meet final position constraint
  - Describes realistic dynamic behaviour

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  - Describes realistic dynamic behaviour

- We need the **'best of both worlds'**

Differential Correction

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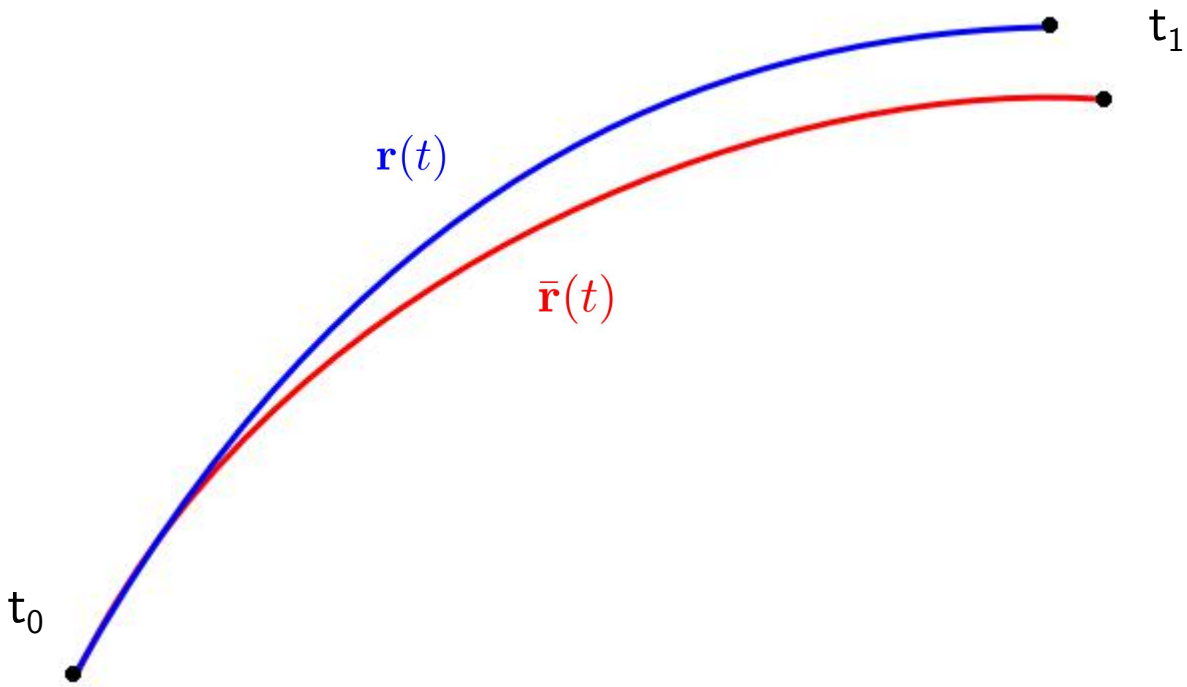
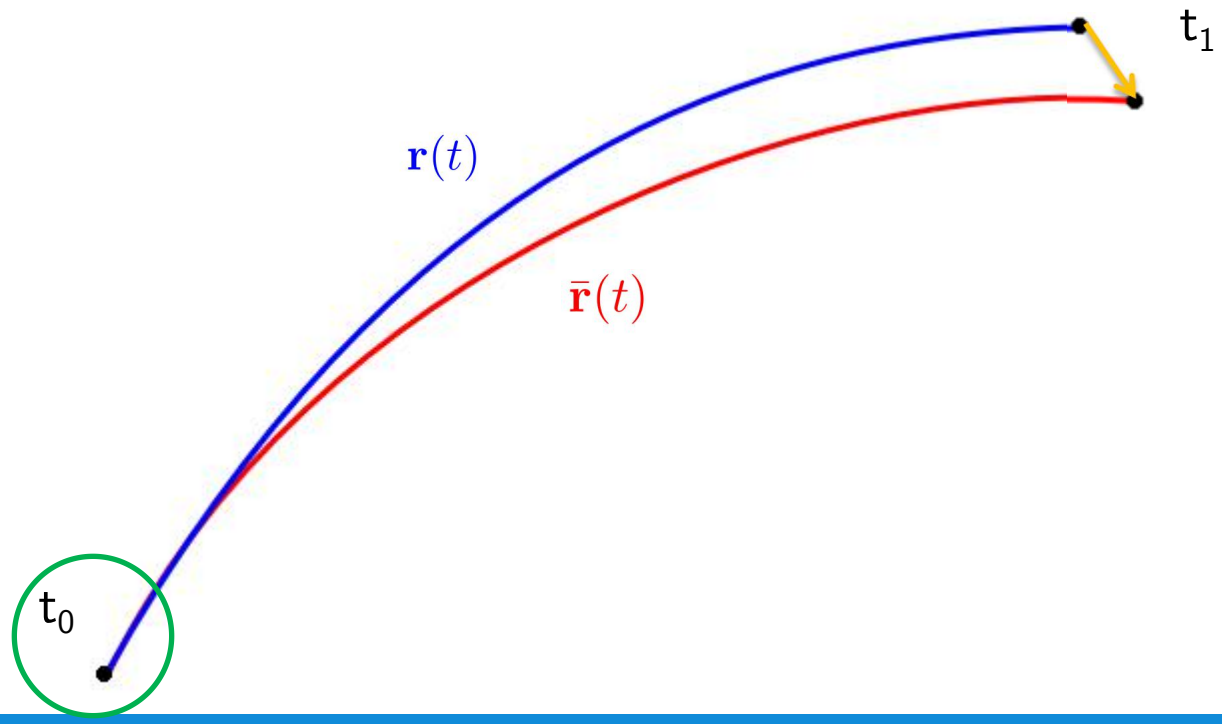


Differential Correction

Introduction

- What we need, **modify the numerical model so that**
  - **Condition** - Meet final position requirement
  - **Condition** - Do not break initial position requirement



Differential Correction

Introduction

- What we need, **modify the numerical model so that**
  - **Condition** - Meet final position requirement
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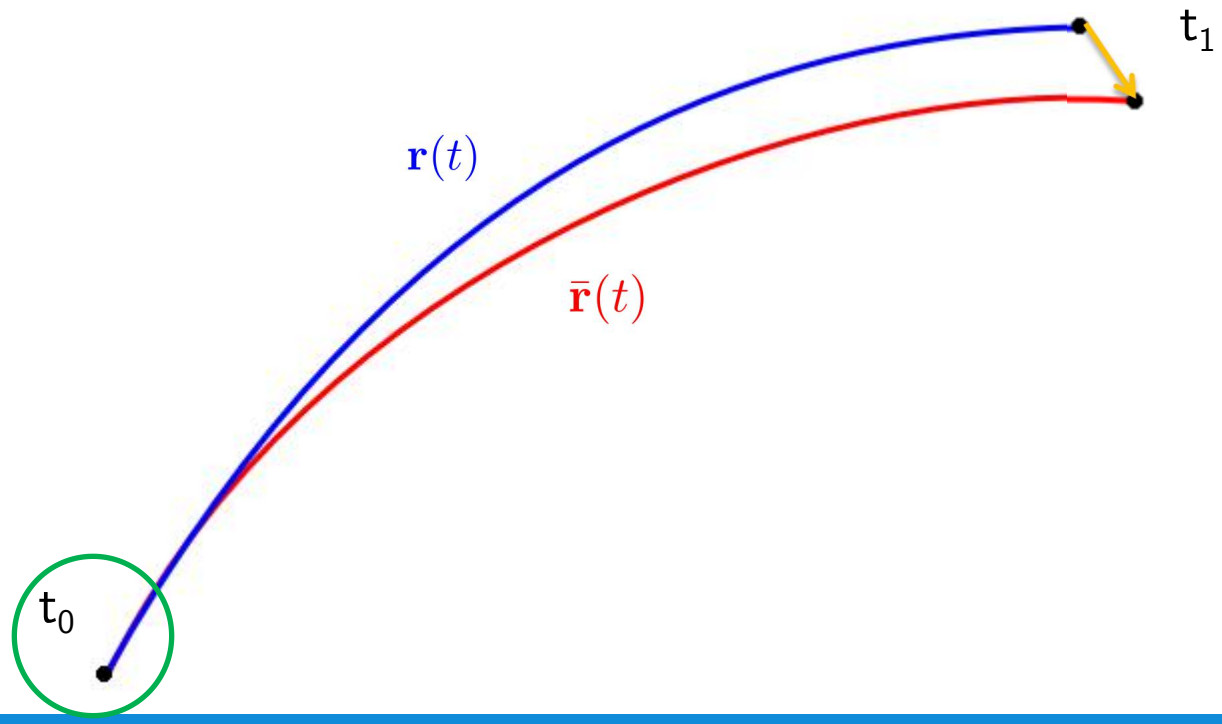
- If the analytical model already meets both conditions, why not use that?
  - Because it is not 'realistic'!



Differential Correction

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- This video

  - Differential correction to use analytical model as 'initial guess'
  - High-thrust - compute single impulsive thrust to meet conditions
  - High-thrust - compute multiple impulsive thrust to meet conditions
  - Low-thrust - compute thrust profile to meet conditions

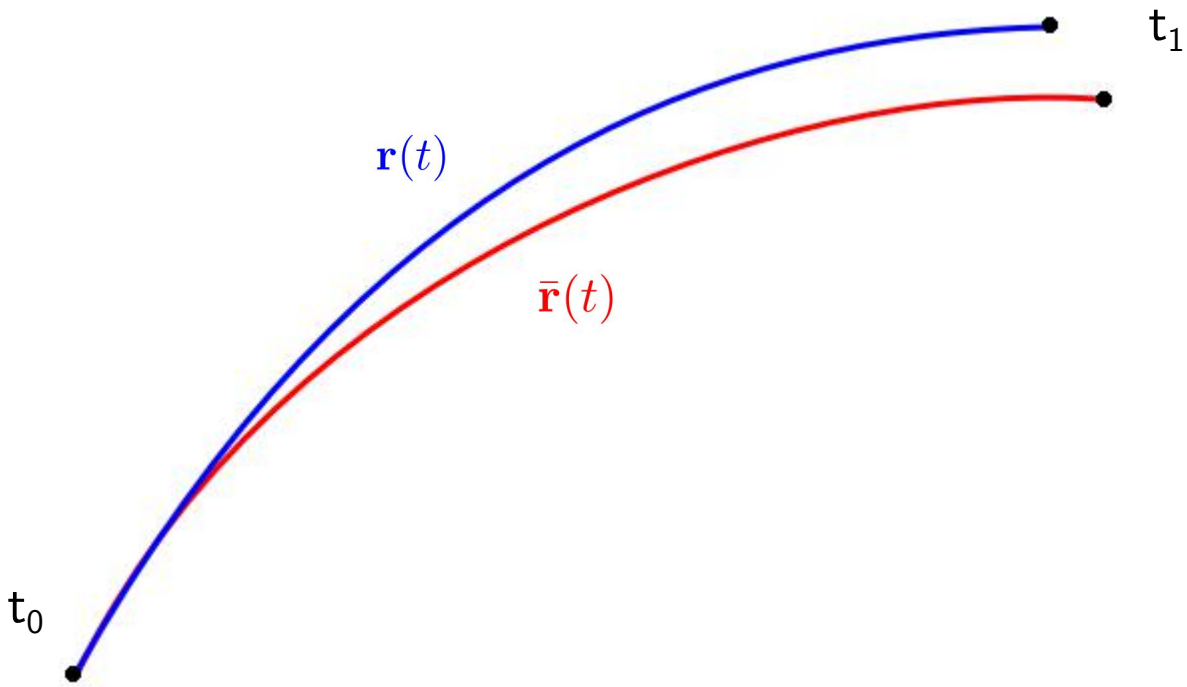
Differential Correction

Defining the objective

- We have a *discrepancy* in the full state (**r** and **v**) at final time t_1
- We have *correspondence* in full state (**r** and **v**) at time t_0

$$\mathbf{r}(t_0) = \bar{\mathbf{r}}(t_0) \qquad \mathbf{v}(t_0) = \bar{\mathbf{v}}(t_0)$$

$$\mathbf{r}(t_1) \neq \bar{\mathbf{r}}(t_1) \qquad \mathbf{v}(t_1) \neq \bar{\mathbf{v}}(t_1)$$



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  - But, we don't need velocity correspondence!
  - Our requirement is only on the positions at t_0 and t_1

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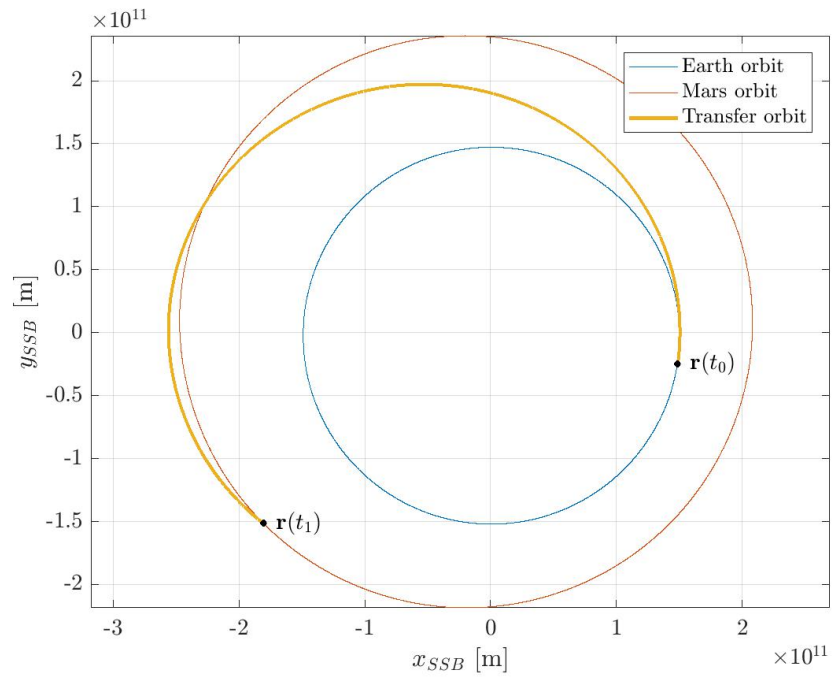

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$$\boxed{\begin{aligned}\mathbf{r}(t_0) &= \bar{\mathbf{r}}(t_0)\\ \mathbf{r}(t_1) &= \bar{\mathbf{r}}(t_1)\end{aligned}} \qquad \boxed{\begin{aligned}\mathbf{v}(t_0) &\neq \bar{\mathbf{v}}(t_0)\\ \mathbf{v}(t_1) &\neq \bar{\mathbf{v}}(t_1)\end{aligned}}$$

- Can we exchange the conditions?
  - Correspondence in both positions
  - Deviations in both velocities

Differential Correction

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Differential Correction

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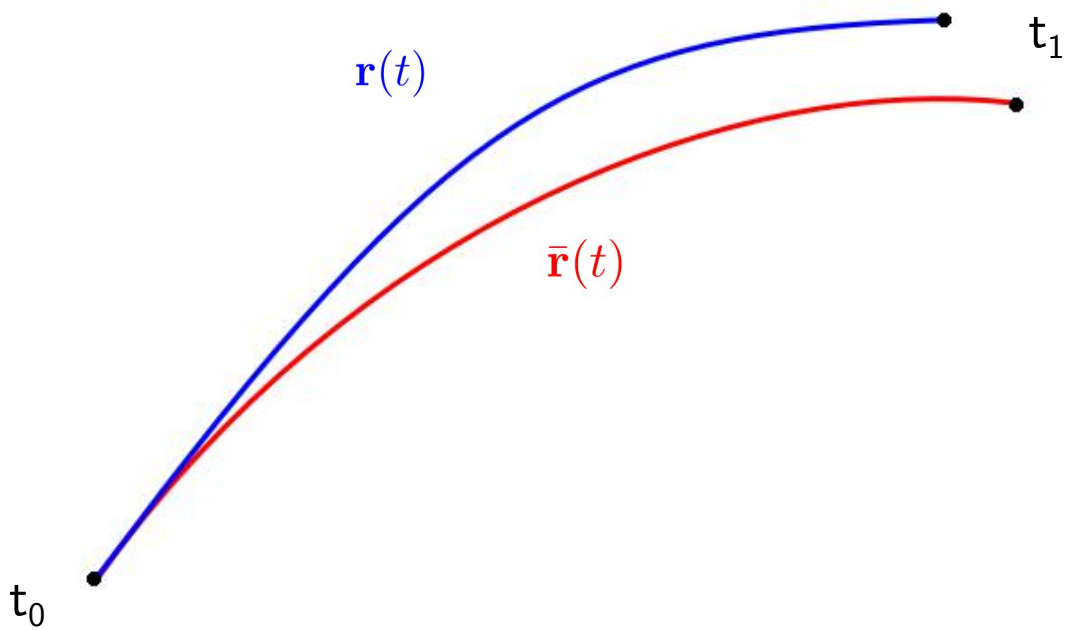

Differential Correction

Reaching the objective - High Thrust

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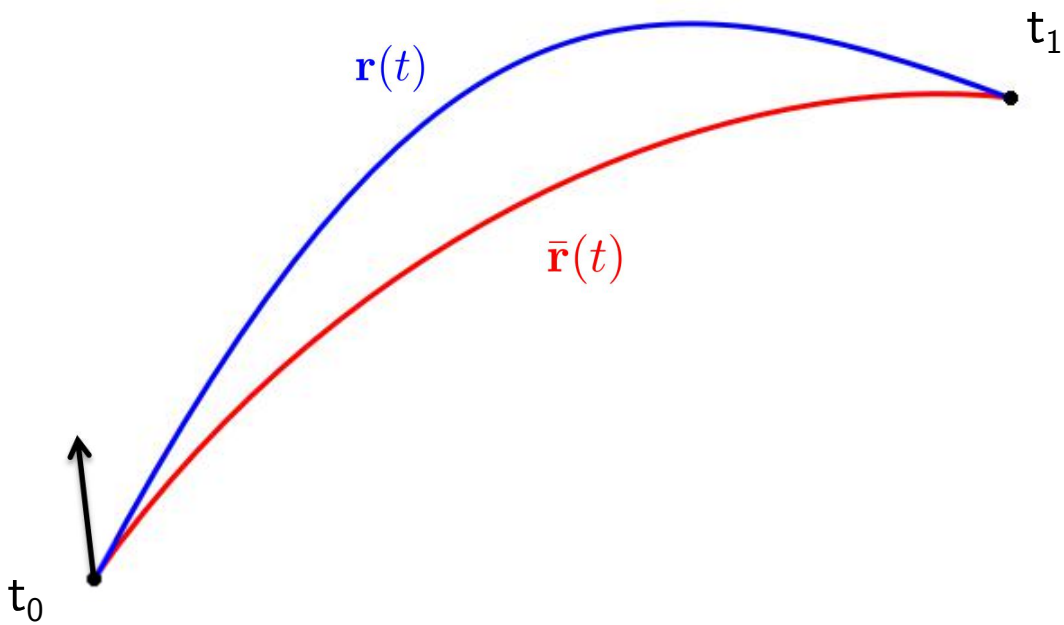
- We can calculate
  - Change in $\mathbf{v}(t_0)$ such that we achieve a required change in $\mathbf{r}(t_1)$



Differential Correction

Reaching the objective - High Thrust

- This is exactly what our state transition matrix can do!

$$\begin{pmatrix} \boxed{\Delta\mathbf{r}(t_1)} \\ \boxed{\Delta\mathbf{v}(t_1)} \end{pmatrix} = \Phi(t_1, t_0) \begin{pmatrix} \boxed{\Delta\mathbf{r}(t_0)} \\ \boxed{\Delta\mathbf{v}(t_0)} \end{pmatrix}$$

- We have
  - Six boundary conditions
  - Six free parameters

$\longrightarrow$ **We can find a unique solution!**

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$\longrightarrow$ **We can find a unique solution!**

- It works **if we assume:**
  - Our 'initial guess' (analytical model, here Lambert arc) is 'close' to the numerical model

Differential Correction

Reaching the objective - High Thrust

- This is exactly what our state transition matrix can do!

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- Procedure:
  - Find $\mathbf{v}(t_0)$ to achieve $\mathbf{r}(t_1)$
  - Calculate $\mathbf{v}(t_1)$ that results from this $\mathbf{v}(t_0)$

$$\Phi(t, t_0) = \begin{pmatrix} \Phi_{rr}(t, t_0) & \Phi_{rv}(t, t_0) \\ \Phi_{vr}(t, t_0) & \Phi_{vv}(t, t_0) \end{pmatrix}$$

$$\Phi_{rr}(t, t_0) = \frac{\partial \mathbf{r}(t)}{\partial \mathbf{r}(t_0)}$$

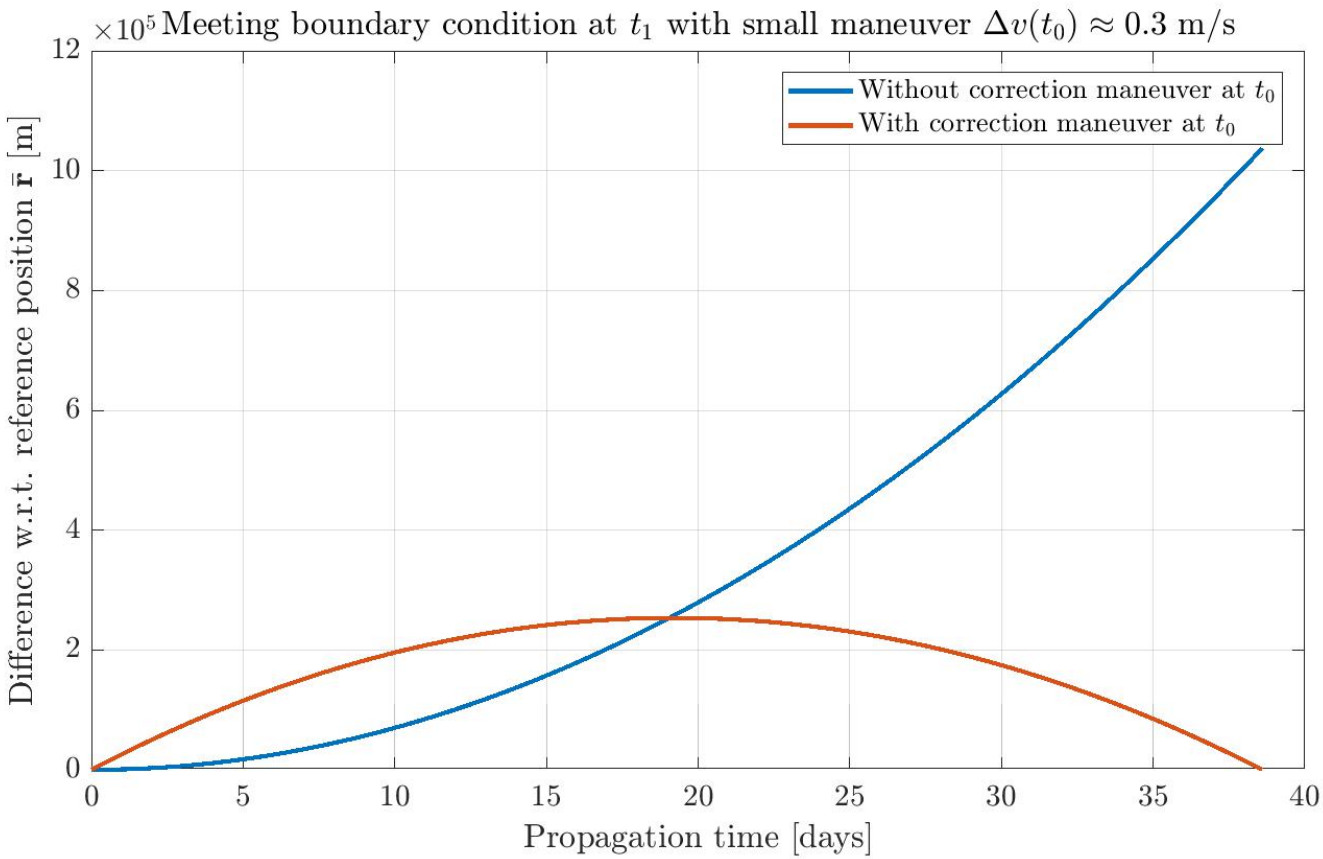
$$\Phi_{rv}(t, t_0) = \frac{\partial \mathbf{r}(t)}{\partial \mathbf{v}(t_0)}$$

$$\ldots$$

Differential Correction

Reaching the objective - High Thrust

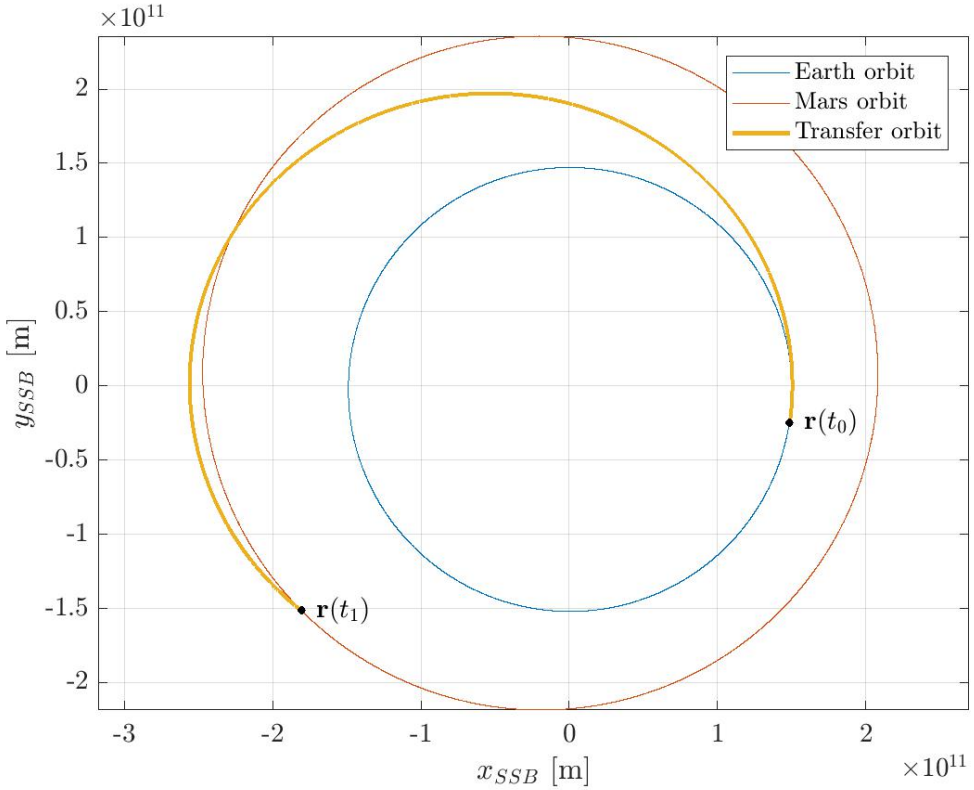
- **Result** - numerically propagated orbit which
  - Meets position constraints at t_0 and t_1
  - Has small velocity deviations w.r.t. reference orbit at t_0 and t_1



Differential Correction

Reaching the objective - High Thrust

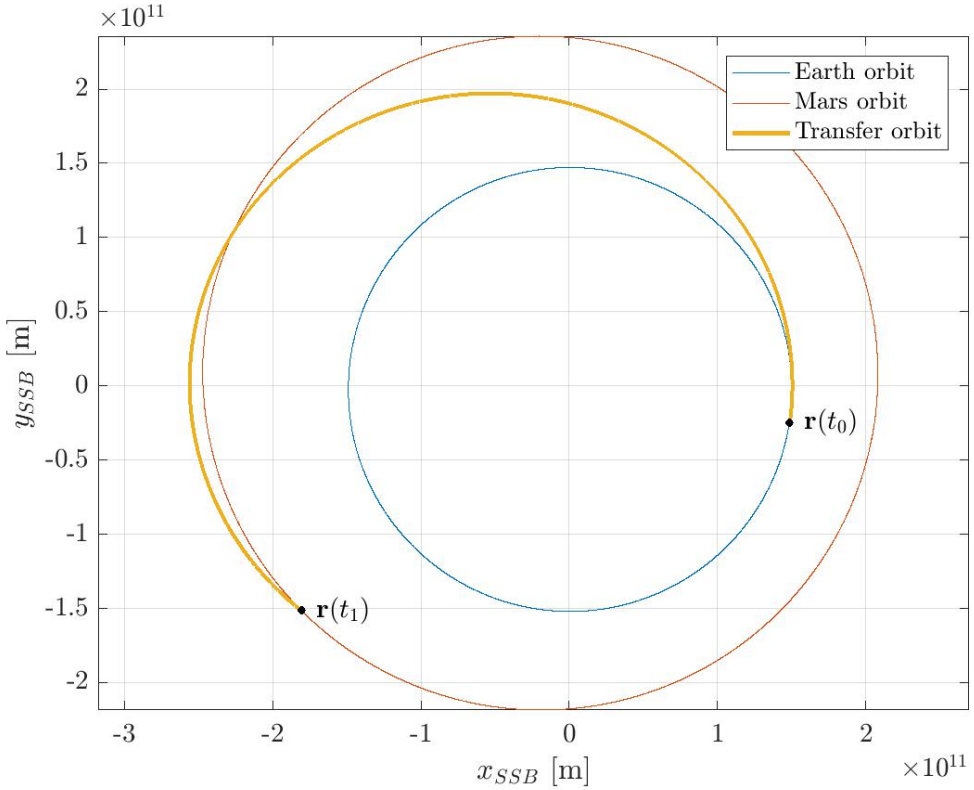
- **For our Lambert arc**
  - Position constraints: depart from Earth, arrive at target planet
  - A large V generally needs to be provided anyway at t_0 and t_1 for escape/capture



Differential Correction

Reaching the objective - Arc-wise

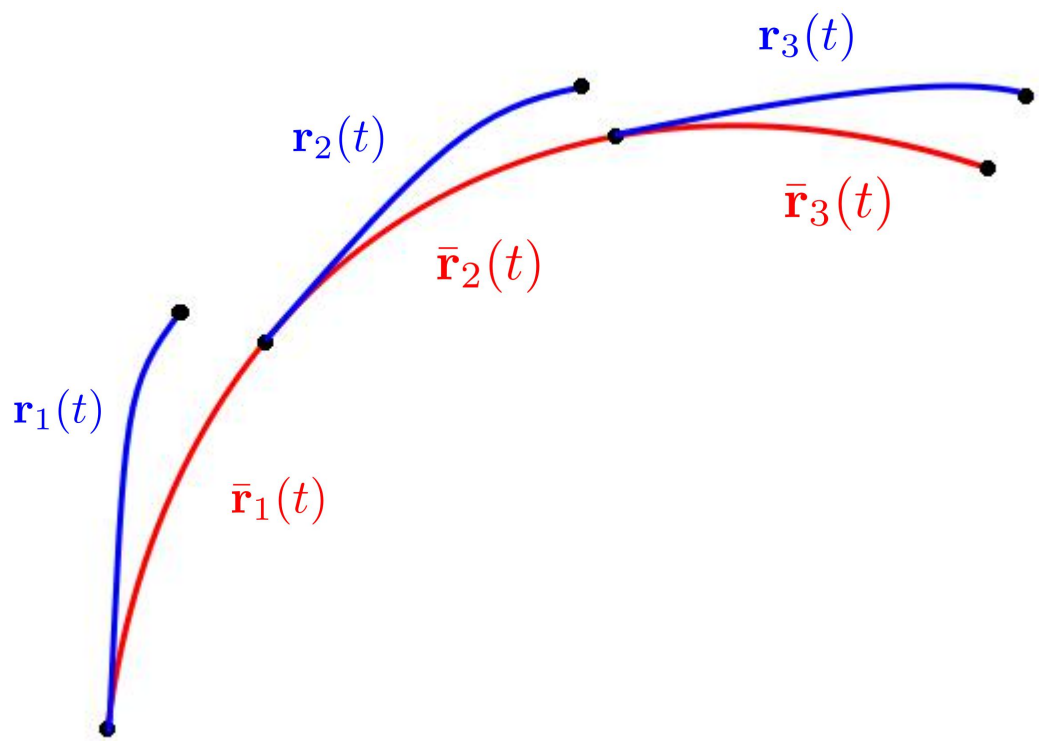
- We assume we do a correction maneuver at t_0 to arrive where we want at t_1
  - **Problem**: t_1-t_0 is large - maneuver 100s of days in advance is not robust (dynamics not predictable enough)!
  - **Solution**: cut our state into arcs!



Differential Correction

Reaching the objective - Arc-wise High-Thrust

- Arc-wise approach
  - Perform *N* separate propagations along the orbit
  - Start the propagation of each arc on our reference orbit



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Reaching the objective - Arc-wise High-Thrust

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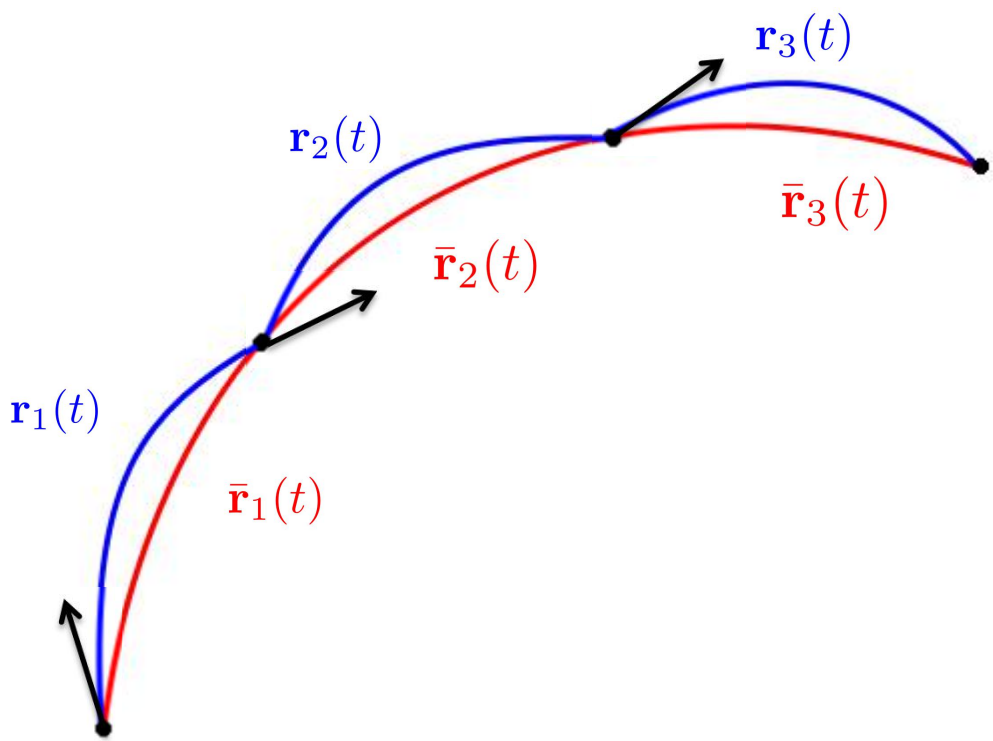
  - Maneuver at start/end of arc: make trajectory continuous in position



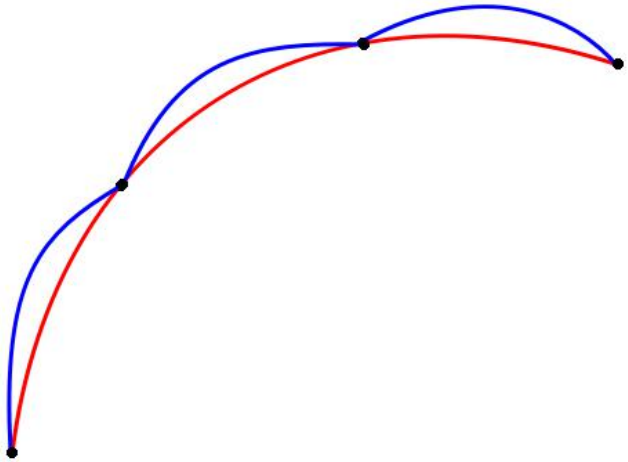
Differential Correction

Reaching the objective - Arc-wise High-Thrust

- Arc-wise approach
  - Perform N separate propagations along the orbit
  - Start the propagation of each arc on our reference orbit
  - Maneuver at start/end of arc: make trajectory continuous in position
- We can still use:

$$\begin{pmatrix} \Delta\mathbf{r}(t_1) \\ \Delta\mathbf{v}(t_1) \end{pmatrix} = \Phi(t_1, t_0) \begin{pmatrix} \Delta\mathbf{r}(t_0) \\ \Delta\mathbf{v}(t_0) \end{pmatrix}$$

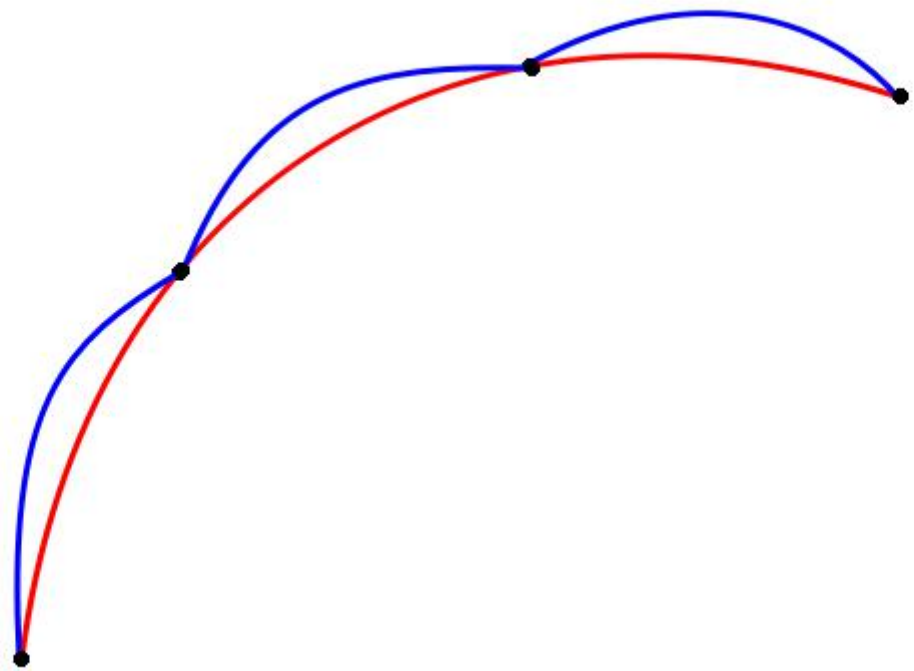
- For each arc!
  - Initial and final position taken from analytical solution



Differential Correction

Reaching the objective - Arc-wise High-Thrust

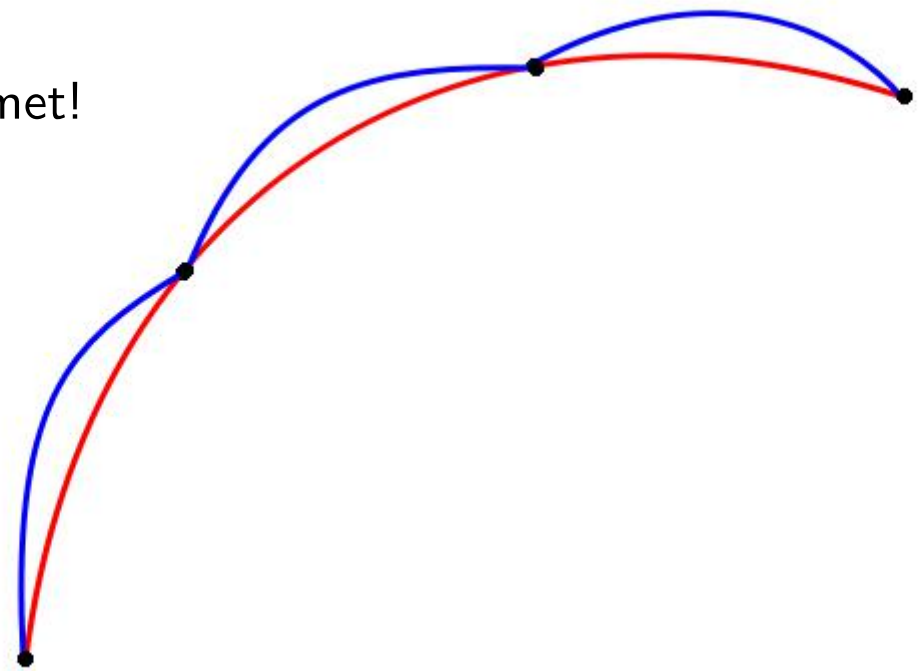
- Arc-wise approach
  - Perform N separate propagations along the orbit
  - **Start the propagation of each arc on our reference orbit**
  - Maneuver at start/end of arc: make trajectory continuous in position
- Why is **this assumption** needed?



Differential Correction

Reaching the objective - Arc-wise High-Thrust

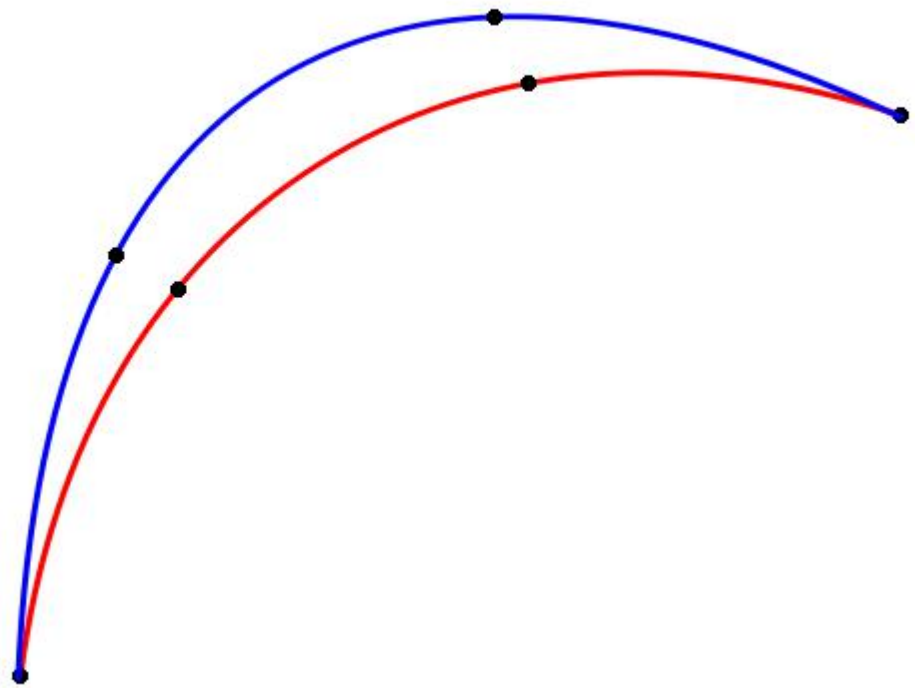
- Arc-wise approach
  - Perform N separate propagations along the orbit
  - **Start the propagation of each arc on our reference orbit**
  - Maneuver at start/end of arc: make trajectory continuous in position
- Why is **this assumption** needed? It's not!
  - Only initial and final position **need** to be met!
  - So why do it?



Differential Correction

Reaching the objective - Arc-wise High-Thrust

- **Consequences**, dropping assumption
  - *Start the propagation of each arc on our reference orbit*



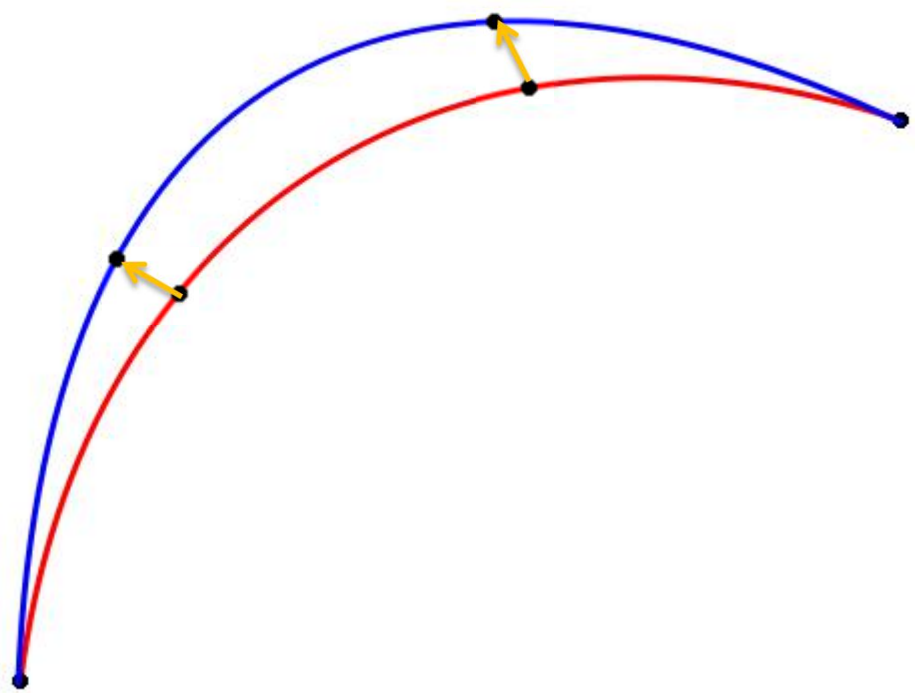
Differential Correction

Reaching the objective - Arc-wise High-Thrust

- **Consequences**, dropping assumption
  - *Start the propagation of each arc on our reference orbit*

- How do we choose the position deviations at intermediate nodes?

  - These are also free parameters!
  - Our system becomes overdetermined



Differential Correction

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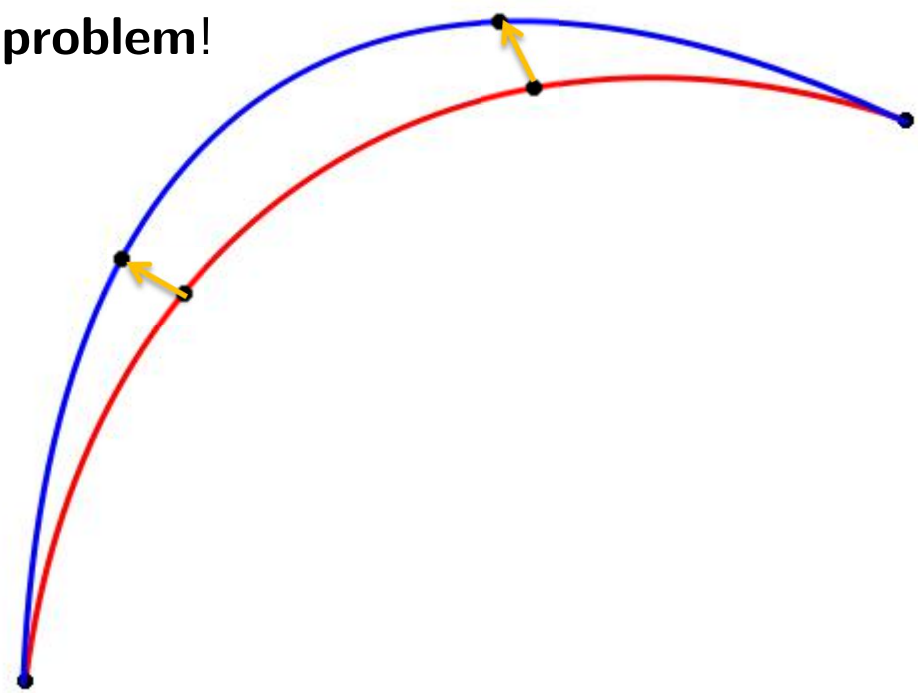
  - We get an **optimization problem**!



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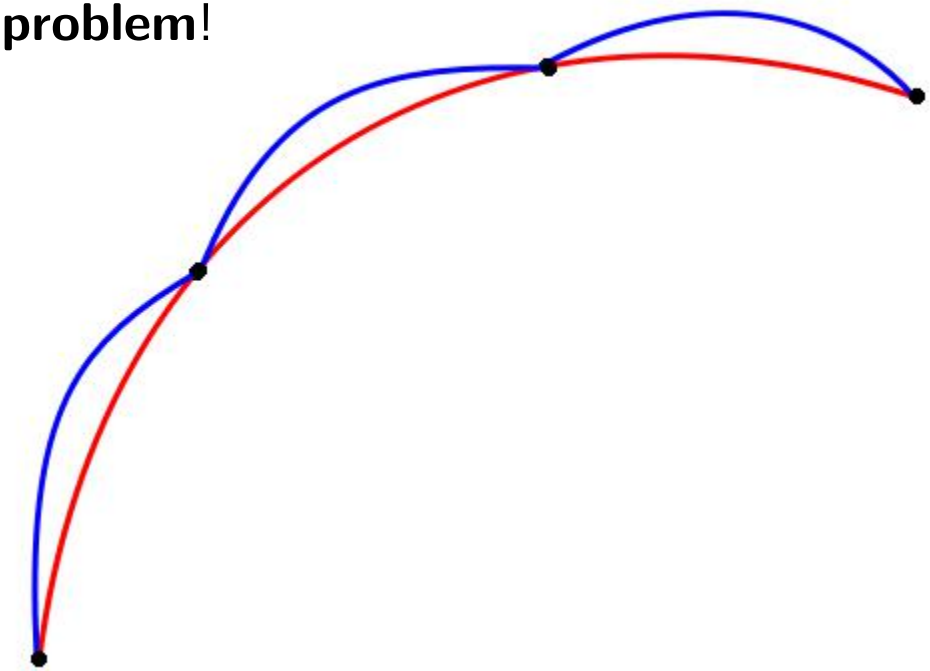
  - Our system becomes overdetermined
  - We get an **optimization problem**!

- Nice, but not in this course :)



Differential Correction

Reaching the objective - Summary

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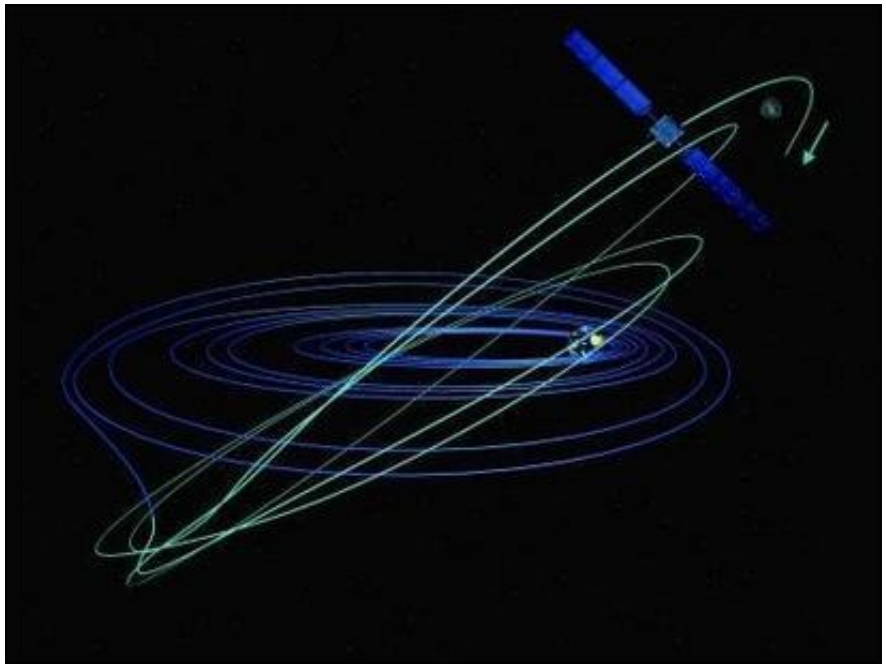
- We need to calculate
  - Change in $\mathbf{v}(t_0)$ such that we achieve a required change in $\mathbf{r}(t_1)$

- Here, we compute an impulsive thrust at beginning (and end) of arc
  - Is this the only approach we can take?
  - No! We can also use a continuous thrust **during** the arc!

Differential Correction

Reaching the objective - Low-Thrust

- Using low-thrust to meet a given final state
  - A very rich and diverse research field
  - Here, we introduce a very basic method



Differential Correction

Reaching the objective - Low-Thrust

- We need to define the **thrust profile**
  - A model for the thrust acceleration $\mathbf{a}_{\mathrm{p}}$ as function of time

- We need a **thrust parameterization**
  - A set of numbers $\mathbf{p}$ that define the behaviour of $\mathbf{a}_{\mathrm{T}}$

Differential Correction

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- We need a **thrust parameterization**
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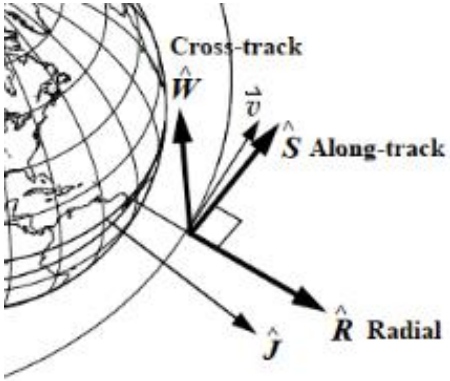
$$\boxed{\mathbf{a}_T = a_R \hat{\mathbf{r}}(t) + a_S \hat{\mathbf{s}}(t) + a_W \hat{\mathbf{w}}(t)}$$

$$\boxed{\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}}$$



Differential Correction

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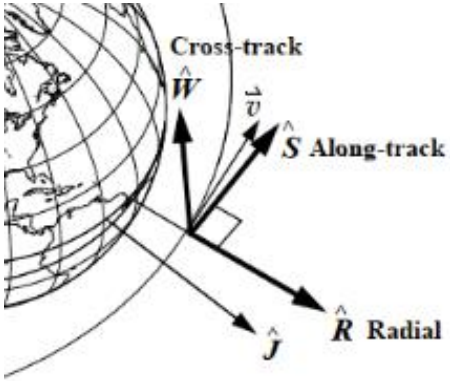

- Note: parameters are constant, unit vector directions are time-dependent

Differential Correction

Reaching the objective - Low-Thrust

- Thrust parameterized by vector **p**
  - Influence of thrust at time t_1

$$\begin{pmatrix} \boxed{\Delta\mathbf{r}(t_1)} \\ \boxed{\Delta\mathbf{v}(t_1)} \end{pmatrix} = \mathbf{S}(t_1)\boxed{\mathbf{p}}$$

  - **Three** boundary conditions
  - **Three** thrust parameters
  - **Three** free velocities

- Solvable: velocity change at t_1 is only an ***output***

Differential Correction

Reaching the objective - Low-Thrust

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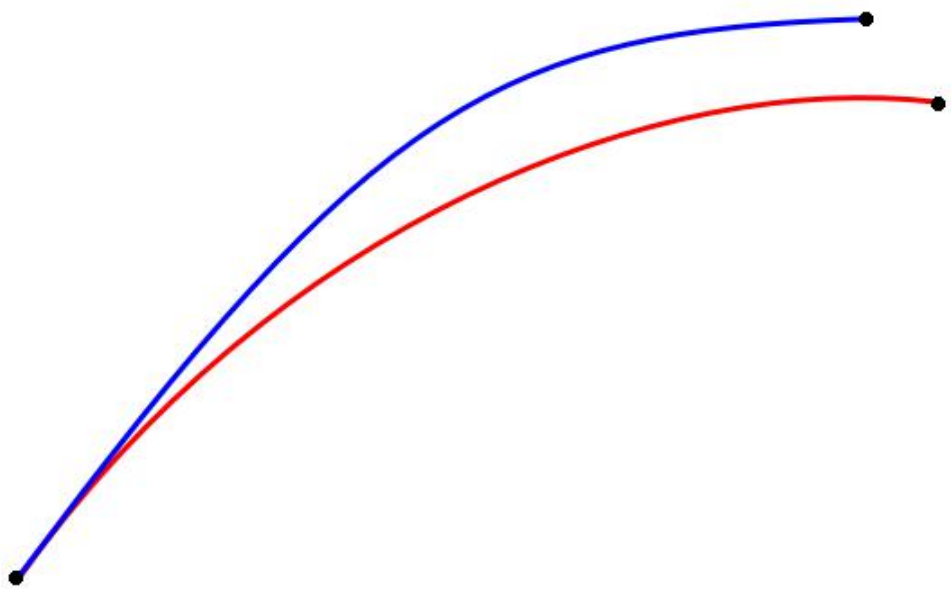
  - Influence of thrust at time t_1

$$\begin{pmatrix} \Delta\mathbf{r}(t_1) \\ \Delta\mathbf{v}(t_1) \end{pmatrix} = \mathbf{S}(t_1)\mathbf{p}$$

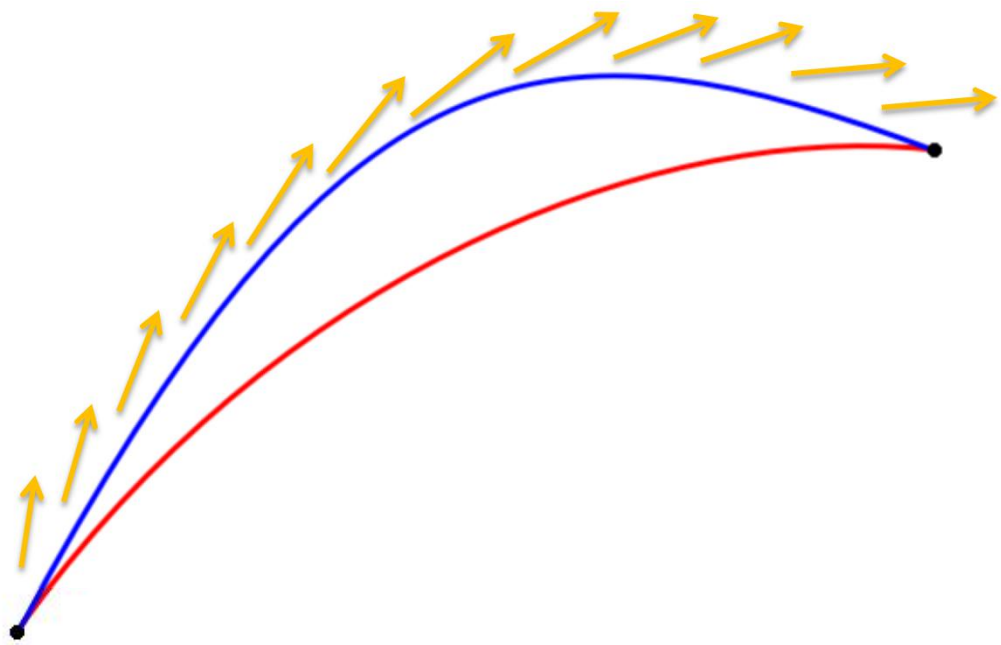


Differential Correction

Reaching the objective - Low-Thrust

- Thrust parameterized by vector **p**
  - Influence of thrust at time t_1

$$\begin{pmatrix} \Delta\mathbf{r}(t_1) \\ \Delta\mathbf{v}(t_1) \end{pmatrix} = \mathbf{S}(t_1)\mathbf{p}$$



Differential Correction

Reaching the objective - Summary

- Let's reflect on what we have, to reach our required $\mathbf{r}(t_1)$:
  - We can change initial state (impulsive burn - total or per arc)

$$\begin{pmatrix} \boxed{\Delta\mathbf{r}(t_1)} \\ \Delta\mathbf{v}(t_1) \end{pmatrix} = \Phi(t_1, t_0) \begin{pmatrix} \Delta\mathbf{r}(t_0) \\ \Delta\mathbf{v}(t_0) \end{pmatrix}$$

Differential Correction

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Differential Correction

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  - Or we can use some low-thrust profile

$$\begin{pmatrix} \boxed{\Delta\mathbf{r}(t_1)} \\ \Delta\mathbf{v}(t_1) \end{pmatrix} = \mathbf{S}(t_1)\mathbf{p} \qquad \mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

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  - In both cases, we end up with a non-zero $\mathbf{v}(t_1)$

Differential Correction

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$$\begin{pmatrix} \boxed{\Delta\mathbf{r}(t_1)} \\ \boxed{\Delta\mathbf{v}(t_1)} \end{pmatrix} = \Phi(t_1, t_0) \begin{pmatrix} \boxed{\Delta\mathbf{r}(t_0)} \\ \Delta\mathbf{v}(t_0) \end{pmatrix} = \mathbf{0}$$

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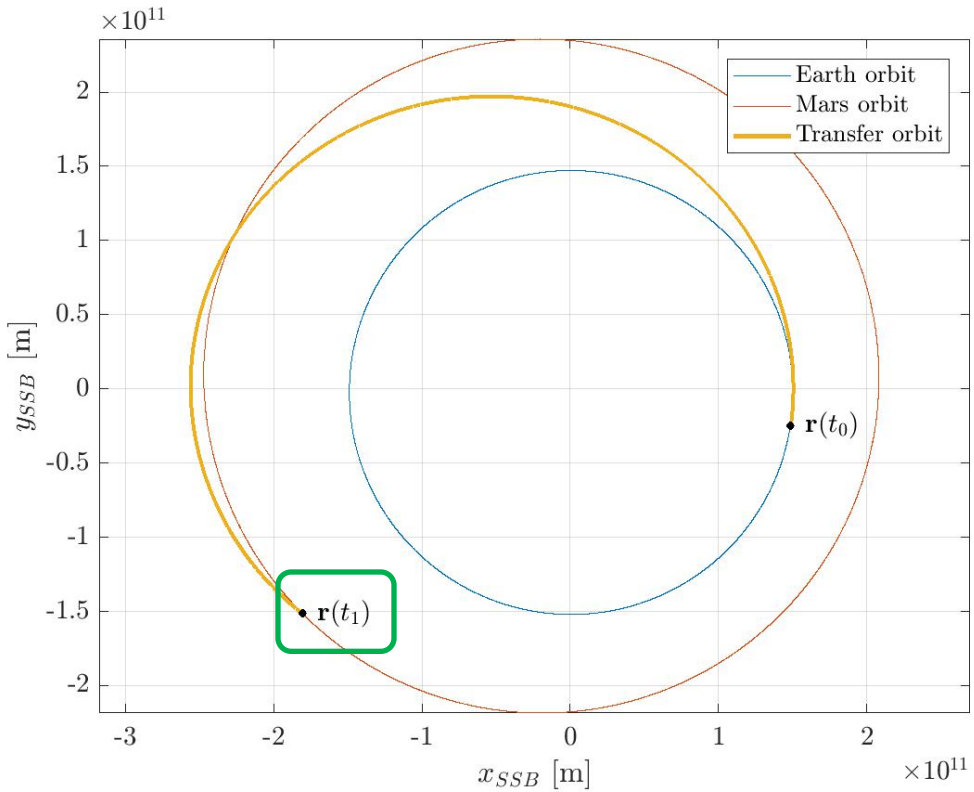
  - In both cases, we end up with a non-zero $\mathbf{v}(t_1)$
    - What if we have a specific required $\mathbf{v}(t_1)$?
    - Additional constraint: the above can no longer be solved!

Differential Correction

Reaching the objective - Position and velocity constraint

- Lambert targeter
  - Only considers interplanetary arc
  - A (large) V will be needed to enter Mars orbit (insertion burn)

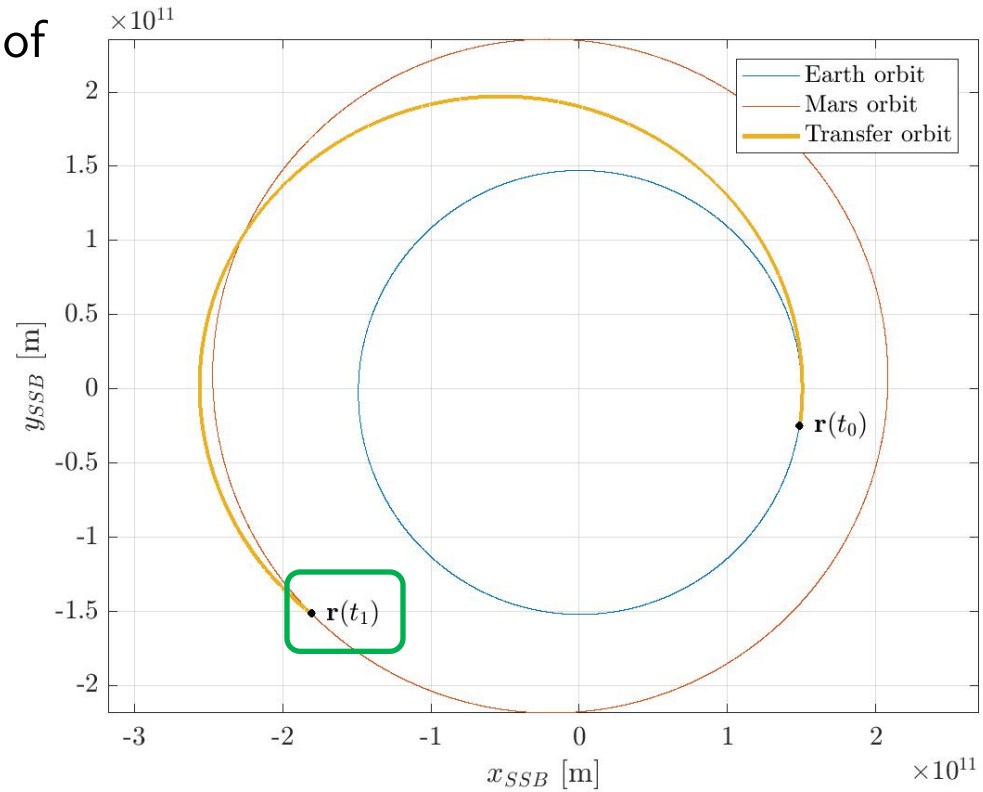


Differential Correction

Reaching the objective - Position and velocity constraint

- Lambert targeter
  - Only considers interplanetary arc
  - A (large) V will be needed to enter Mars orbit (insertion burn)

- What about Low-thrust?
  - We may not be capable of insertion burn
  - Then: we have to arrive at Mars **with a specific velocity**

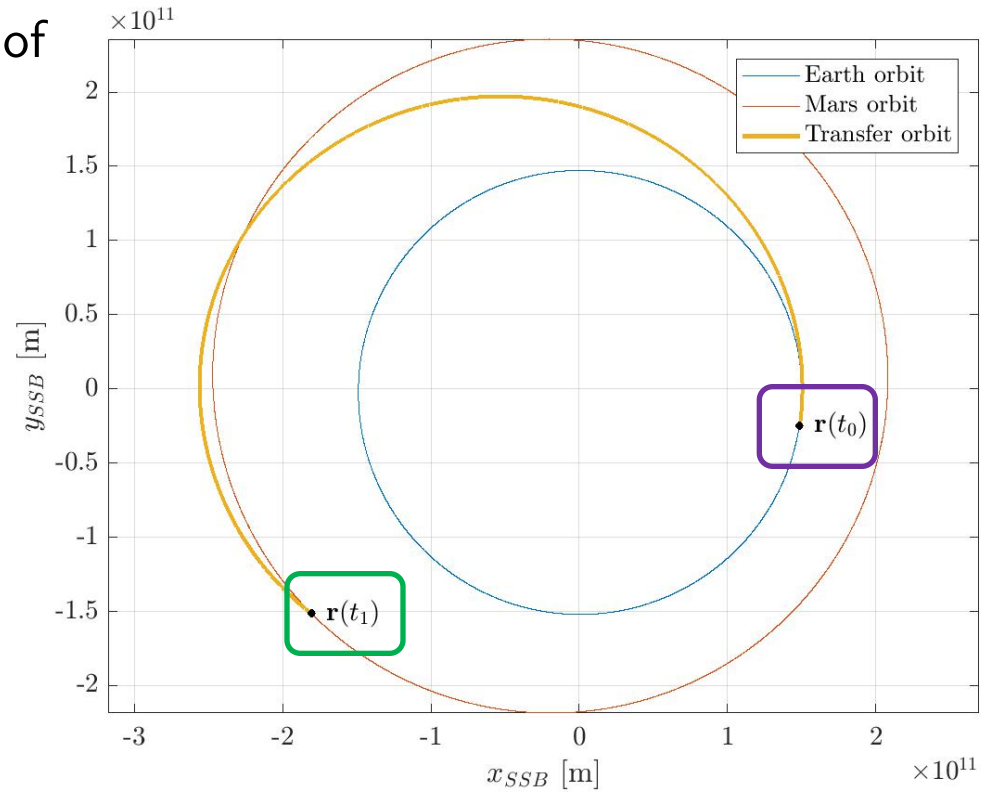


Differential Correction

Reaching the objective - Position and velocity constraint

- Lambert targeter
  - Only considers interplanetary arc
  - A (large) V will be needed to enter Mars orbit (insertion burn)

- What about Low-thrust?
  - We may not be capable of insertion burn
  - Then: we have to arrive at Mars **with a specific velocity**
  - (same may apply for escaping Earth orbit)



Differential Correction

Reaching the objective - Position and velocity constraint

- To solve:

$$\begin{pmatrix} \Delta\mathbf{r}(t_1) \\ \Delta\mathbf{v}(t_1) \end{pmatrix} = \mathbf{S}(t_1)\mathbf{p}$$

$$\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

- With constraints on both **r** and **v** at t_1
  - We need more than three free parameters!

- Possible approach
  - Divide thrust into 'arcs'

Differential Correction

Reaching the objective - Position and velocity constraint

- To solve:

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- With constraints on both **r** and **v** at t_1
  - We need more than three free parameters!

- Possible approach
  - Divide thrust into 'arcs'

- Same story as before?
  - Yes and no :)

Differential Correction

Reaching the objective - Arc-wise Low-thrust

- To solve:

$$\begin{pmatrix} \Delta\mathbf{r}(t_1) \\ \Delta\mathbf{v}(t_1) \end{pmatrix} = \mathbf{S}(t_1)\mathbf{p}$$

$$\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

- Arc-wise low-thrust model
  - Constant RSW thrust on each arc

$$\mathbf{a}_T = a_R\hat{\mathbf{r}}(t) + a_S\hat{\mathbf{s}}(t) + a_W\hat{\mathbf{w}}(t)$$

Differential Correction

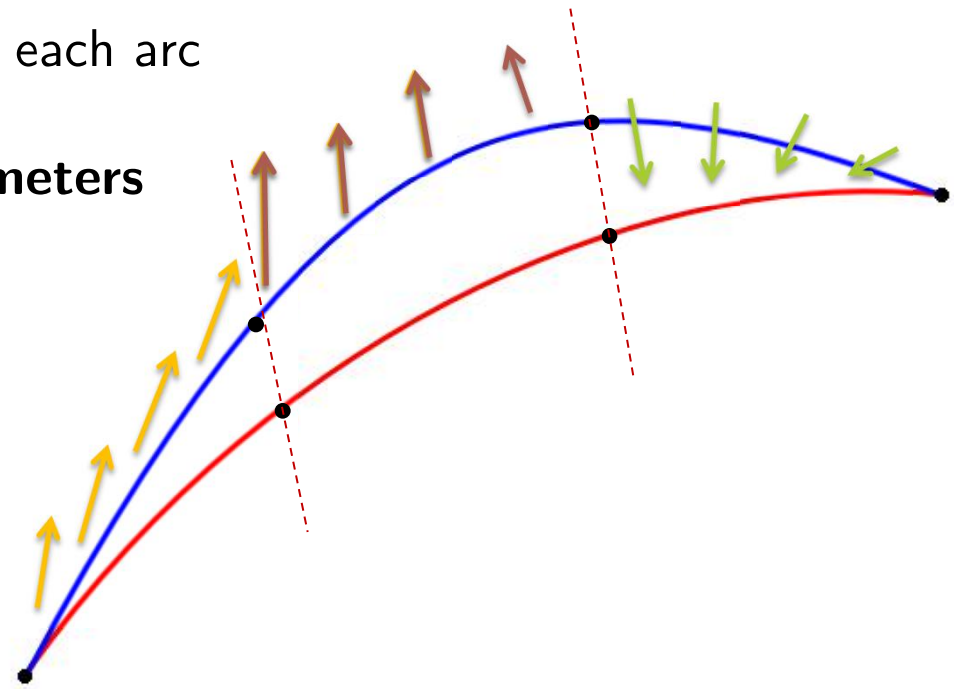
Reaching the objective - Arc-wise Low-thrust

- To solve:

$$\begin{pmatrix} \Delta\mathbf{r}(t_1) \\ \Delta\mathbf{v}(t_1) \end{pmatrix} = \mathbf{S}(t_1)\mathbf{p}$$

$$\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

- Arc-wise low-thrust model
  - Constant RSW thrust on each arc
  - Example: **three arcs**
  - Total of **nine free parameters**



Differential Correction

Reaching the objective - Arc-wise Low-thrust

- To solve:

$$\begin{pmatrix} \Delta\mathbf{r}(t_1) \\ \Delta\mathbf{v}(t_1) \end{pmatrix} = \mathbf{S}(t_1)\mathbf{p}$$

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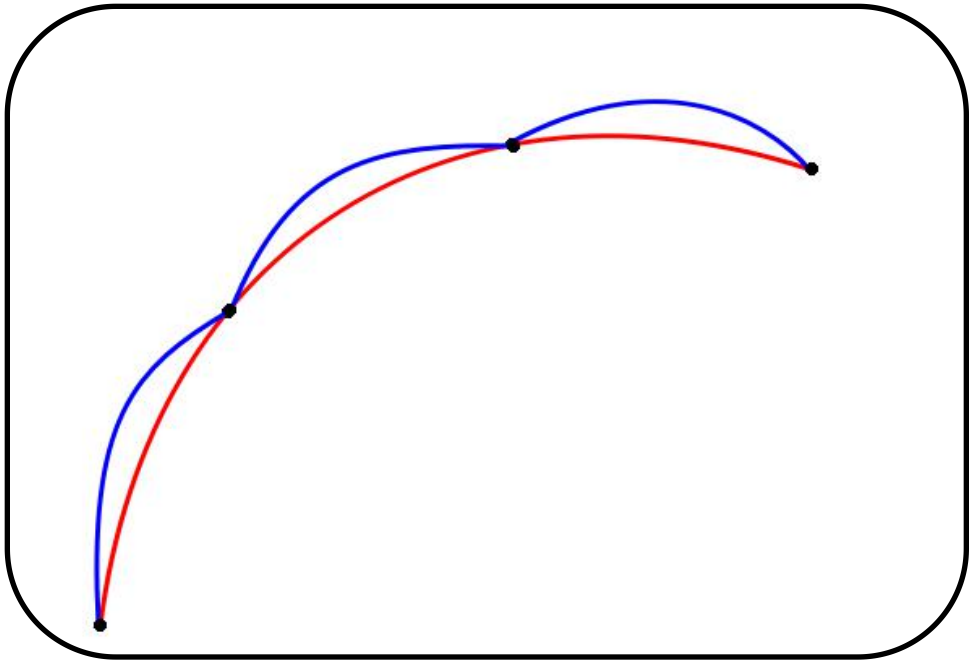
- Arc-wise low-thrust model
  - Constant RSW thrust on each arc

- For high-thrust, **we had:**



Differential Correction

Reaching the objective - Arc-wise Low-thrust

- To solve:

$$\begin{pmatrix} \Delta\mathbf{r}(t_1) \\ \Delta\mathbf{v}(t_1) \end{pmatrix} = \mathbf{S}(t_1)\mathbf{p}$$

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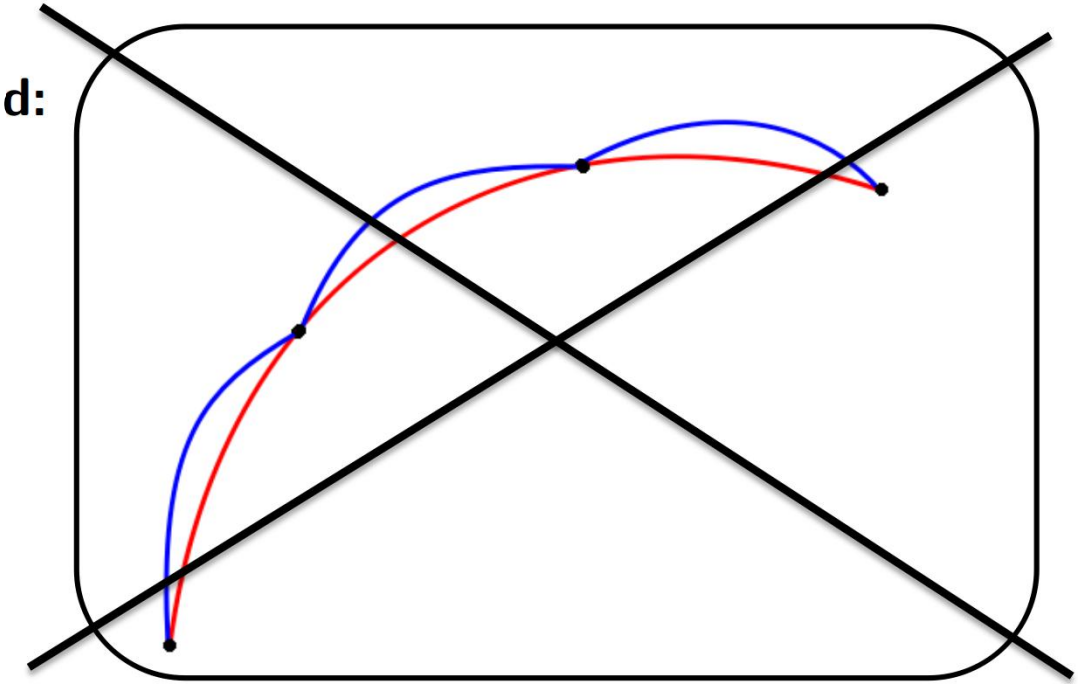
- Arc-wise low-thrust model
  - Constant RSW thrust on each arc
- For high-thrust, **we had:**
- Not for low-thrust!
  - Velocity must be continuous



Differential Correction

Reaching the objective - Arc-wise Low-thrust

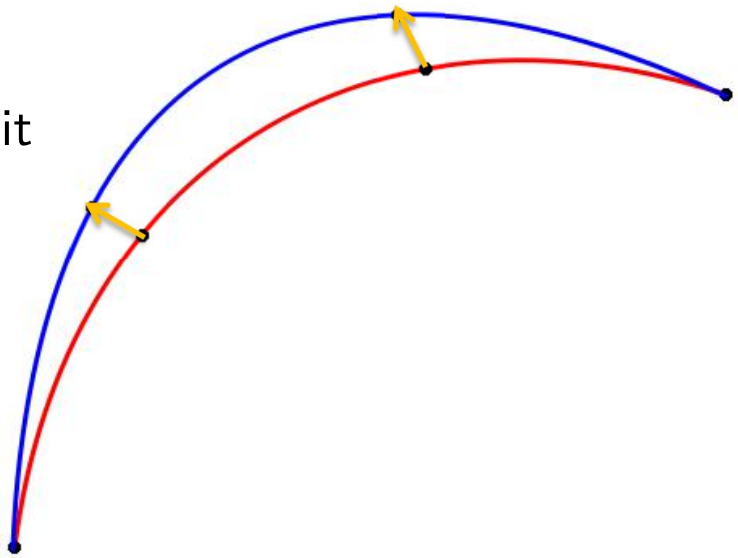
- To solve:

$$\begin{pmatrix} \Delta\mathbf{r}(t_1) \\ \Delta\mathbf{v}(t_1) \end{pmatrix} = \mathbf{S}(t_1)\mathbf{p}$$

$$\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

- Arc-wise low-thrust model
  - Constant RSW thrust on each arc

- Solution:
  - Drop requirement: having arc start/end positions **on** reference orbit
  - Add requirement: velocity must be continuous between two arcs



Differential Correction

Reaching the objective - Arc-wise Low-thrust

$$\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

- Divide thrust into 'arcs'
  - Use a thrust parameterization of 3 parameters for each arc

Differential Correction

Reaching the objective - Arc-wise Low-thrust

$$\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

- Divide thrust into 'arcs'
  - Use a thrust parameterization of 3 parameters for each arc
- For two arcs
  - Requirements: 6 (final **position and velocity** of **full** trajectory)
  - Free parameters: 6 (for each arc: 3 thrust parameters)
  - Exactly solvable

Differential Correction

Reaching the objective - Arc-wise Low-thrust

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Differential Correction

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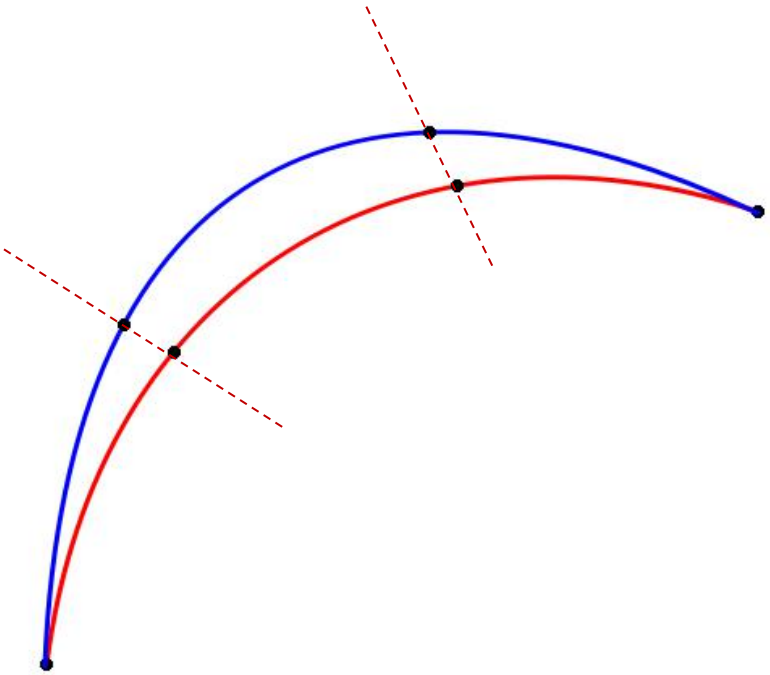
- Divide thrust into 'arcs'
  - Use a thrust parameterization of 3 parameters for each arc
- For two arcs
  - Requirements: 6 (final **position and velocity** of **full** trajectory)
  - Free parameters: 6 (for each arc: 3 thrust parameters)
  - Exactly solvable
- For *N* arcs
  - Requirements: 6 (initial and final position of **full** trajectory)
  - Free parameters: 3N (for each arc: 3 thrust parameters)
  - Infinite number of solutions for N>2 → optimization problem!

Differential Correction

Reaching the objective - Arc-wise Low-thrust

$$\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

- Divide thrust into 'arcs'
  - How to compute the behaviour?



Differential Correction

Reaching the objective - Arc-wise Low-thrust

$$\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

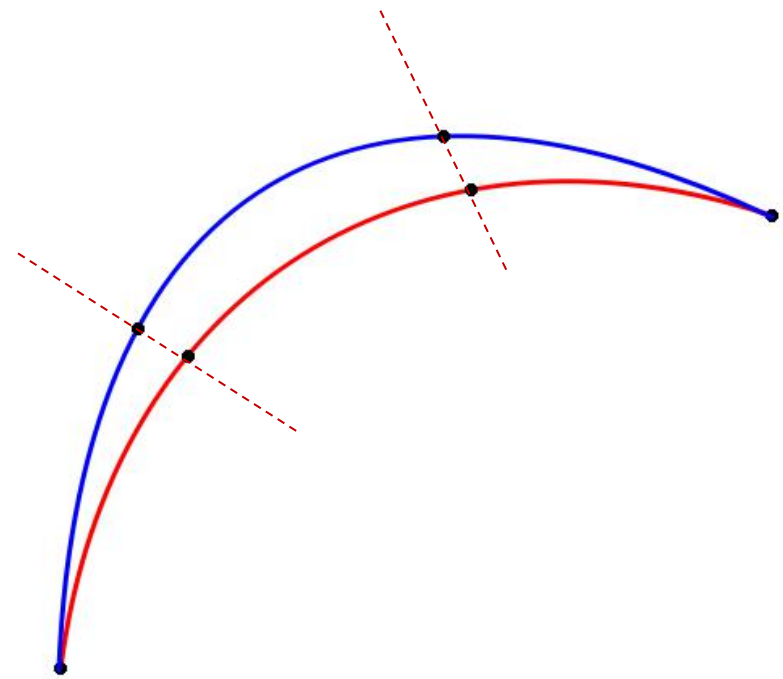
- Divide thrust into 'arcs'
  - How to compute the behaviour?

- For each arc:

$$\Delta\mathbf{x}_i(t_{E,i}) = \Phi_i(t_{E,i}, t_{0,i})\Delta\mathbf{x}_i(t_{0,i}) + \mathbf{S}_i(t_{E,i})\mathbf{p}_i$$

- Notation:

$t_{0,i}$	Initial time: arc i
$t_{E,i}$	Final time: arc i
$\mathbf{x}_i$	State: arc i
$_i(t,t_{0,i})$	State transition mat., arc i
$\mathbf{S}_i(t)$	Sensitivity mat., arc i



Differential Correction

Reaching the objective - Arc-wise Low-thrust

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  - How to compute the behaviour?

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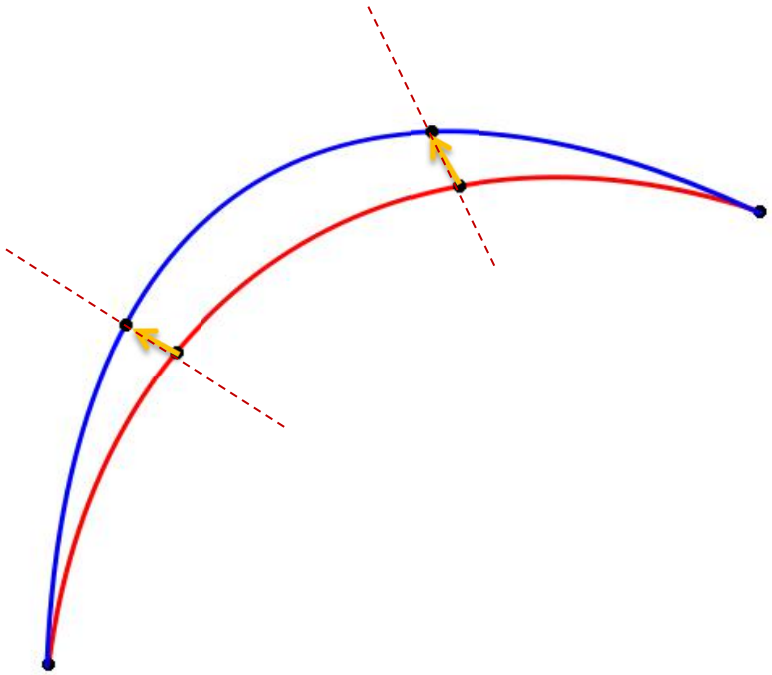
- Constraint between arcs:

$$\mathbf{x}_i(t_{E,i}) = \mathbf{x}_{i+1}(t_{0,i})$$



Differential Correction

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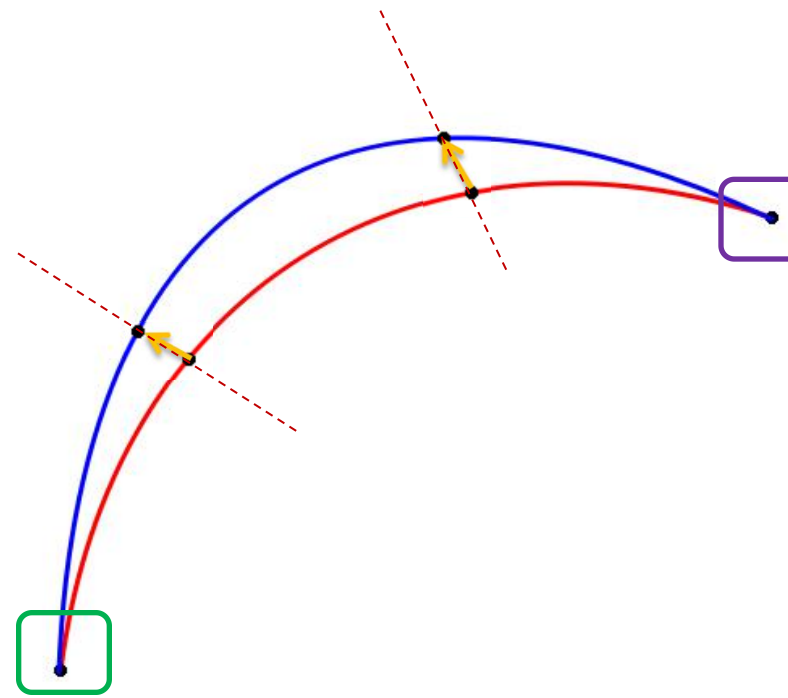
- Constraint between arcs:

$$\mathbf{x}_i(t_{E,i}) = \mathbf{x}_{i+1}(t_{0,i})$$

- Boundary conditions:

$$\mathbf{x}_0(t_{0,i}) = \mathbf{x}_0$$

$$\mathbf{x}_N(t_{E,N}) = \mathbf{x}_E$$



Differential Correction

Reaching the objective - Arc-wise Low-thrust

$$\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

- Divide thrust into 'arcs'
  - How to compute the behaviour

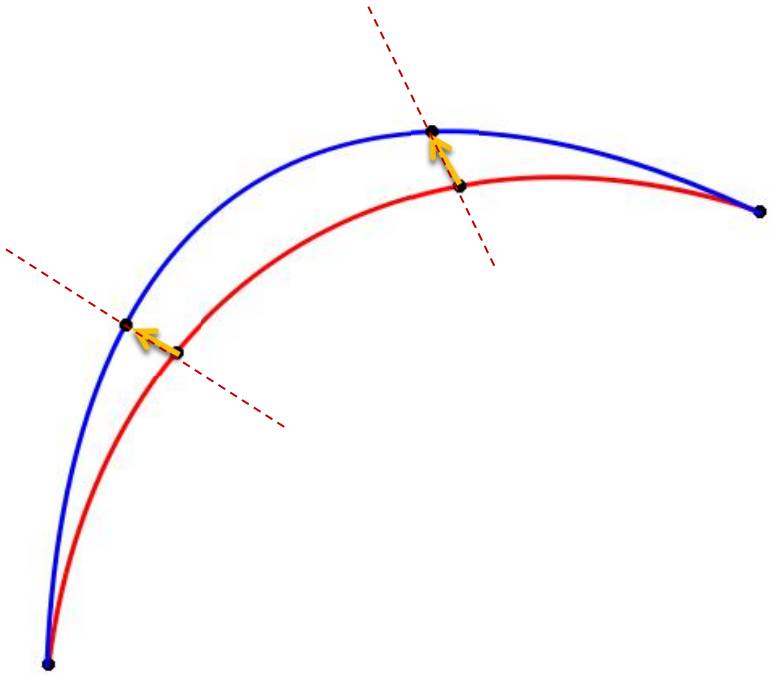
$$\Delta \mathbf{x}_i(t_{E,i}) = \Phi_i(t_{E,i}, t_{0,i}) \Delta \mathbf{x}_i(t_{0,i}) + \mathbf{S}_i(t_{E,i}) \mathbf{p}_i$$

$$\mathbf{x}_i(t_{E,i}) = \mathbf{x}_{i+1}(t_{0,i})$$

$$\mathbf{x}_0(t_{0,i}) = \mathbf{x}_0$$

$$\mathbf{x}_N(t_{E,N}) = \mathbf{x}_E$$

- Optimization problem!
  - Find values of $\mathbf{p}_i$
  - Meet constraints
  - **Minimize propellant**



Differential Correction

Reaching the objective - Arc-wise Low-thrust

$$\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

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$$\mathbf{x}_i(t_{E,i}) = \mathbf{x}_{i+1}(t_{0,i})$$

$$\boxed{\begin{aligned} \mathbf{x}_0(t_{0,i}) &= \mathbf{x}_0 \\ \mathbf{x}_N(t_{E,N}) &= \mathbf{x}_E \end{aligned}}$$

- Specific case (N=2)
  - **Full constraint on initial and final state**
  - Above has a unique solution

Differential Correction

Reaching the objective - Arc-wise Low-thrust

$$\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

- Divide thrust into 'arcs'
  - How to compute the behaviour

$$\Delta \mathbf{x}_i(t_{E,i}) = \Phi_i(t_{E,i}, t_{0,i}) \Delta \mathbf{x}_i(t_{0,i}) + \mathbf{S}_i(t_{E,i}) \mathbf{p}_i$$

$$\mathbf{x}_i(t_{E,i}) = \mathbf{x}_{i+1}(t_{0,i})$$

$$\mathbf{r}_0(t_{0,i}) = \mathbf{r}_0$$

$$\mathbf{r}_N(t_{E,N}) = \mathbf{r}_E$$

  - The whole algorithm works just as well for other constraints!

Differential Correction

Reaching the objective - Arc-wise Low-thrust

$$\mathbf{p} = \begin{pmatrix} a_R \\ a_S \\ a_W \end{pmatrix}$$

- Divide thrust into 'arcs'
  - How to compute the behaviour

$$\Delta \mathbf{x}_i(t_{E,i}) = \Phi_i(t_{E,i}, t_{0,i}) \Delta \mathbf{x}_i(t_{0,i}) + \mathbf{S}_i(t_{E,i}) \mathbf{p}_i$$

$$\mathbf{x}_i(t_{E,i}) = \mathbf{x}_{i+1}(t_{0,i+1})$$

$$\mathbf{r}_0(t_{0,i}) = \mathbf{r}_0$$

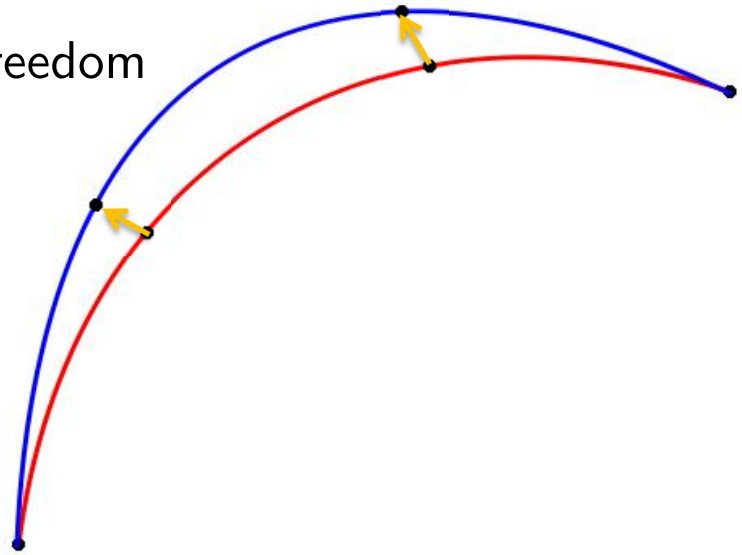
$$\mathbf{r}_N(t_{E,N}) = \mathbf{r}_E$$

- Specific case (N=2)
  - **Constraint on initial and final position only**
  - Possible approach
    - Guess thrust profile for arc 1
    - Compute thrust profile for arc 2
    - Redo for multiple guesses to 'find optimum'

Differential Correction

Reaching the objective - Arc-wise Low-Thrust Summary

- Meeting velocity constraints at end of trajectory
  - Requires more freedom in thrust design

- For low-thrust
  - Introduce more free parameters
  - Do not enforce absolute constraints on arc interfaces
  - Introduce continuity constraints

- Arc-wise thrust definition
  - Allows arbitrary amount of model freedom
  - Generally: optimization problem